In [78]:
```r
# Generate 100 random numbers from Uniform Distribution
x <- runif(100, min = 50, max = 100)

# Generate 100 random numbers from Normal Distribution
y <- rnorm(100, mean = 70, sd = 70)

print(x)
print(y)
```

```
  [1] 71.66761 97.99338 86.80779 57.51158 62.83510 73.87702 61.63514
70.94519
  [9] 77.96703 56.83759 71.97736 53.48813 84.41277 51.09132 92.66251
83.64676
 [17] 66.83522 84.69953 80.26182 57.88257 86.79503 63.82006 64.59666
62.17887
 [25] 77.72704 86.23645 95.34702 94.39766 77.57057 50.35175 84.42735
90.19025
 [33] 66.87960 76.43277 66.16485 98.84647 81.27668 57.66362 69.18993
70.26101
 [41] 83.10775 92.59634 60.11716 56.03461 81.45647 65.38507 91.85133
88.90398
 [49] 88.06724 59.66556 50.34645 68.46320 54.10403 77.92197 51.71266
71.72402
 [57] 96.75533 55.02197 55.63459 87.60952 60.21735 88.90839 95.12694
66.42585
 [65] 75.85922 79.51152 66.83603 74.78489 82.86281 66.31720 74.56211
96.30018
 [73] 85.78355 58.06526 69.54829 89.40535 60.85344 80.18199 62.33296
54.96467
 [81] 77.90573 52.69684 71.83633 77.10238 92.54769 79.14861 90.54268
96.93288
 [89] 75.84347 74.96852 88.88214 74.94743 58.48357 78.46544 94.18863
51.85300
 [97] 80.23169 50.33729 93.34727 60.55677
  [1]   77.422935  181.344402   68.722871   54.752394  170.298312  1
48.296356
  [7]  119.248397   -3.009495  102.843733  126.964534   78.947040  -
16.273494
 [13]  -24.044667  -58.873611   61.901367  223.664175  126.644833
38.024499
 [19]   42.022721  115.883812   30.332255   61.051786    5.210749  1
21.282119
 [25]  279.398097  142.686902   47.394082  102.669534  144.152602
66.186617
 [31]  104.127680  -50.833894  149.607743  234.891629   79.650544
46.835763
 [37]   14.156854  107.068292   88.578887  -81.045430  -31.586677
72.289529
 [43]   43.567476  109.677653   80.313174   48.486288  -12.222052  -
32.161519
 [49]   95.685893  188.777085  138.824963   18.782736   68.565168
81.445487
 [55]   85.004974   22.397846   66.578636  153.432158   26.466732
10.298658
 [61]  152.725660   56.809195   67.076530   24.996309  -32.472956
```

```
25.366710
 [67]   88.127331   27.014022  124.275780  138.862539   82.913217
73.692932
 [73]    3.614061   53.212278  191.263923   69.280315 -126.178250  2
23.701748
 [79]   28.620593   45.628307   43.350686   -6.615659   54.167984
80.196328
 [85]   85.774185   83.132463   82.443628   40.348021   44.218884
16.836984
 [91]    3.737401   78.973250  -17.556212  -16.076555   79.944124  1
12.839550
 [97]  -22.924871   87.242389   54.876170  225.371080
```

In [79]:
```
cat("Uniform Distribution\n")
cat("Mean:", mean(x), "\n")
cat("Median:", median(x), "\n\n")

cat("Normal Distribution\n")
cat("Mean:", mean(y), "\n")
cat("Median:", median(y), "\n")
```

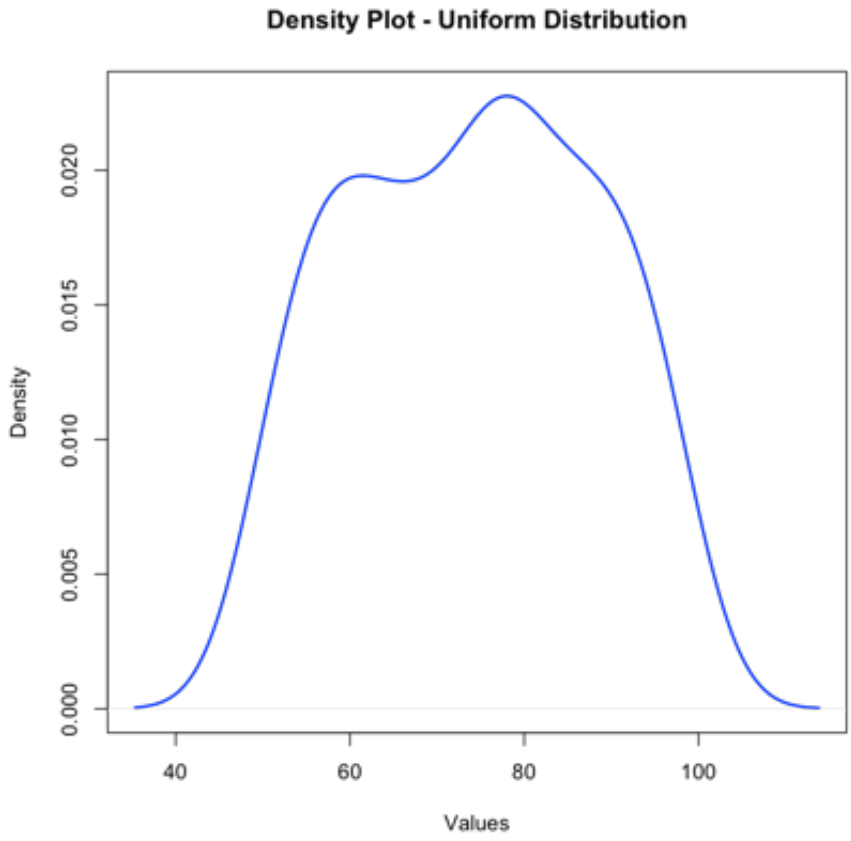
```
Uniform Distribution
Mean: 74.11535
Median: 74.95798

Normal Distribution
Mean: 69.91618
Median: 68.64402
```

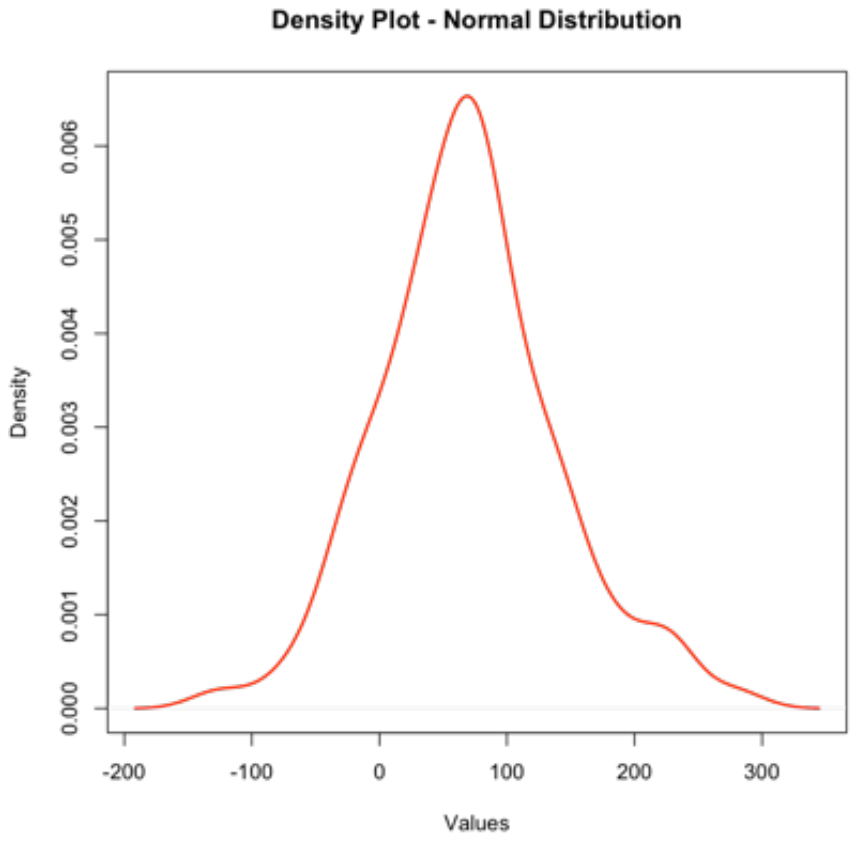
In [80]:
```
table_x <- table(x)
names(table_x)[which.max(table_x)]
```

'50.3372855950147'

In [81]:
```
plot(
  density(x),
  main = "Density Plot - Uniform Distribution",
  xlab = "Values",
  ylab = "Density",
  col = "blue",
  lwd = 2
)
```

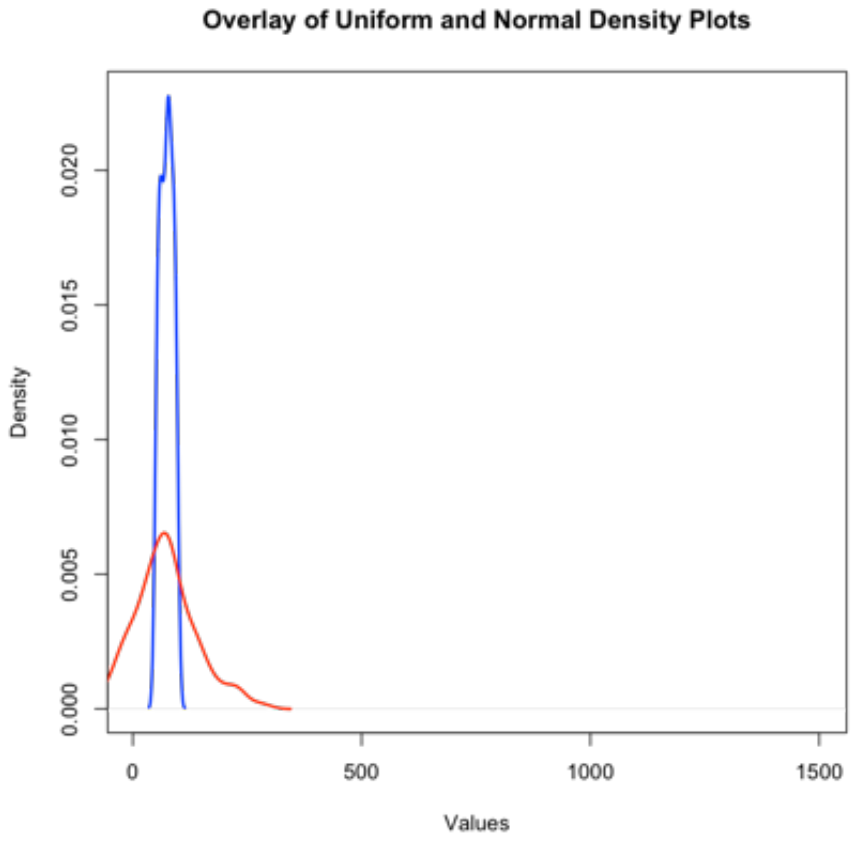



In [82]:
```
plot(density(y),
  main = "Density Plot – Normal Distribution",
  xlab = "Values",
  ylab = "Density",
  col = "red",
  lwd = 2
)
```




In [83]:
```
plot(
  density(x),
  col = "blue",
  lwd = 2,
  xlim = c(5,1500),
  main = "Overlay of Uniform and Normal Density Plots",
  xlab = "Values",
  ylab = "Density"
)

lines( density(y),col = "red",lwd = 2)
```




In [84]:
```
x <- runif(1000, 50, 100)
z <- runif(1000, 50, 10000)
```

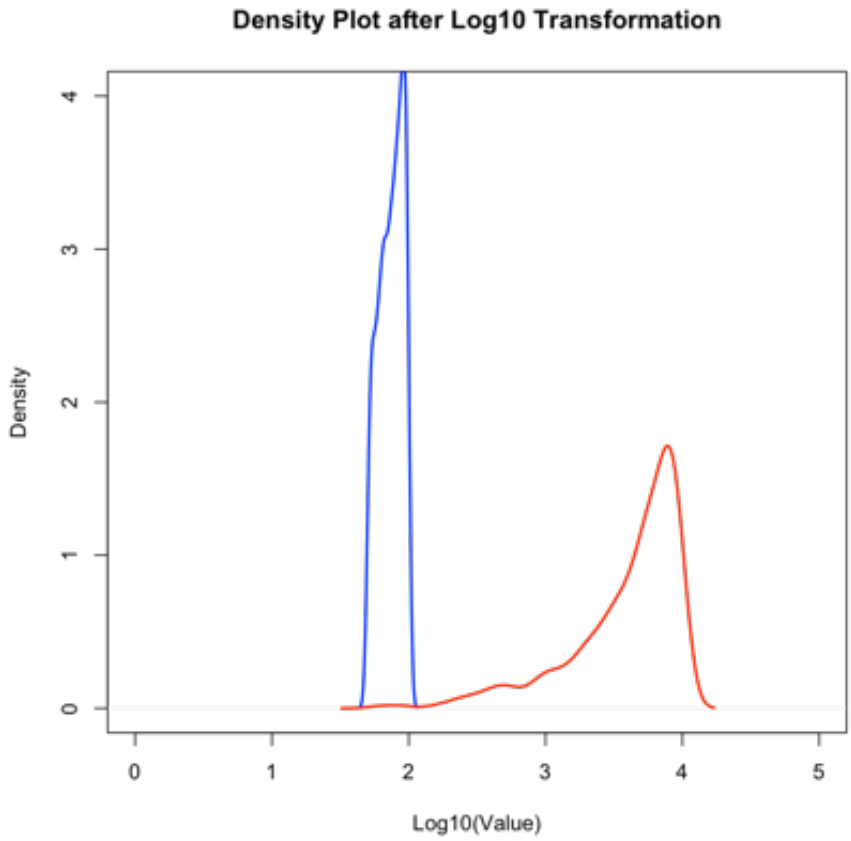
In [85]:
```
log_x <- log10(x)

log_z <- log10(z)
```

In [86]:
```
d1 <- density(log_x)
d2 <- density(log_z)

plot(
  d1,
  xlim = range(c(0,5)),
  ylim = range(c(0,4)),
  col = "blue",
  lwd = 2,
  main = "Density Plot after Log10 Transformation",
  xlab = "Log10(Value)",
  ylab = "Density"
)

lines(d2, col = "red", lwd = 2)
```


Density Plot after Log10 Transformation
Density
0
1
2
3
4
0
1
2
3
4
5
Log10(Value)

In [ ]:

In [ ]:

In [ ]:

In [ ]: